

Section B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6.	(a)	(i)		$\Delta E = hf$ and $f = c/\lambda$ (h = Planck's constant, c = speed of light) (1) $\text{energy} = hc/\lambda = 6.63 \times 10^{-34} \times 3.00 \times 10^8 / 656 \times 10^{-9}$ (1) $\text{energy} = 3.03 \times 10^{-19} \text{ J}$ (1)		3		3	3	
		(ii)		measuring the convergent frequency / wavelength at the convergence limit (1) calculate the ionisation energy using $\Delta E = hf$ (1)	2			2		
	(b)	(i)		hydrogen lower value since it has a smaller nuclear charge (1) electron comes from the same shell / no extra shielding (1)	2			2		
		(ii)		award (1) for any two of following - hydrogen higher value since no shielding of outer electron / lithium has shielding of outer electron - outweighs smaller nuclear charge / greater effective nuclear charge - H outer electron closer to nucleus / lithium loses electron from higher energy sub-shell	2			2		

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	(c)	(i)		$V_2 = \frac{P_1 V_1 T_2}{P_2 T_1}$ (1) $V_2 = \frac{1.01 \times 10^5 \times 125 \times 325}{1.01 \times 10^5 \times 297}$ $V_2 = 137 \text{ cm}^3$ (1)	1	1		2	2	
		(ii)		$n = \frac{PV}{RT}$ (1) $V = 1.6 \times 10^{-4} \text{ m}^3$, $T = 293 \text{ K}$ (1) $n = \frac{1.01 \times 10^5 \times 1.6 \times 10^{-4}}{8.31 \times 293}$ $n = 6.64 \times 10^{-3} \text{ mol}$ (1)	1	2		3	3	
	(d)			the student is incorrect because: H_2O contains 2 bonding pairs and 2 lone pairs of electrons (1) BeH_2 contains 2 bonding pairs only (1) therefore different shapes since electron pairs arrange themselves to be as far apart as possible/ different number of electron pairs (1)			3	3		

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	(e)	(i)		volume is 50 cm ³ therefore mass is 50 g (1) density = 50/54.5 = 0.917 g cm ⁻³ (1)		2		2	1	
		(ii)		any of following for (1) - hydrogen bonding exists in both water and ice but it extends throughout the whole structure in ice - in ice hydrogen bonds hold the molecules together in an open (tetrahedral) structure - the molecules are further apart in ice than they are in water (so ice is less dense than water)	1			1		
				Question 6 total	9	8	3	20	9	0